

SCHOOLPULSE

School Management System

Business Plan — Version 1.0 — March 2026

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01

Executive Summary

The opportunity, the product, the numbers

SchoolPulse is a cloud-based School Management SaaS purpose-built for Kenyan primary and secondary schools. It digitises fee collection, attendance, academic records, parent communication, and discipline management into one affordable platform accessible from any browser or phone.

Kenya has over 39,000 schools. Government audits have confirmed KES 3.7 billion lost to ghost students, 87,000 fraudulent enrollments, and fee mismanagement that has driven schools to closure. The dominant tools are paper registers and WhatsApp groups. SchoolPulse replaces them.

The opportunity: 8,500+ private and low-cost private schools, zero credible affordable alternatives, and a payment rail (M-Pesa) that makes frictionless digital fee collection immediately possible.

Metric	Value
Total Kenyan schools (primary + secondary)	~39,000
Primary target: private & low-cost private schools	8,500+
Government audit loss (2020–2024)	KES 3.7 billion to ghost students
KCSE pass rate (C+ and above, 5-year avg)	21.07%
Monthly subscription range	KES 7,500 – 38,000+ per term
Break-even (paying schools)	2 schools on Growth tier
Year 3 target ARR	KES 42,000,000+
Founder background	Full-stack (React, Node.js, Java, JavaFX)
Go-live strategy	2-school pilot (1 primary, 1 secondary) — Term 2 2026

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The Problem

Evidence-backed. Real schools. Real losses.

2.1 Ghost Students & Fraudulent Enrollment

 Government Audit (2020–2024): 87,000 ghost learners found in 723 of 1,039 sampled schools. KES 3.59 billion overpaid in capitation. Three schools shared one bank account receiving KES 103.3 million fraudulently. Root cause: paper registers with zero audit trail.

2.2 Schools That Failed Due to Poor Management

School	Outcome	Root Cause
Kiambogo Secondary School	Closed January 2025	Insurmountable debt from untracked fee collection
St. Peter’s Barichu Secondary	Near-closure	KCSE collapse, no academic tracking or early intervention
Gabondu North Secondary, Kirinyaga	Forced closure risk	Poor admission records, no management system
Mbea Secondary School	Performance degradation	Chronic untracked truancy, no intervention mechanism

2.3 KCSE Performance Crisis

Only 21.07% of all KCSE candidates between 2020–2024 scored C+ or above (909,784 of 4,317,349). Schools that consistently top rankings — Alliance, Kabarak, Kapsabet — all have structured internal management systems. The gap is not intelligence. It is infrastructure.

2.4 Communication & Transparency Gap

- Parents learn of fee arrears only at the school gate on reporting day
- Dominant communication channel is WhatsApp — unstructured, unverifiable, inaccessible to feature-phone users
- Report cards delivered once per term on paper — if they arrive at all
- Ministry of Education has repeatedly flagged undisclosed fees and certificate withholding

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The Solution

SchoolPulse — what it does and why it wins

3.1 Core Product

SchoolPulse is a multi-tenant SaaS platform with six core modules available to every subscriber. Each module addresses a specific documented failure point in Kenyan school management.

Module	What It Solves	Killer Feature
Student Register	No central student record, manual admission tracking	Soft-delete audit trail, bulk CSV import, multi-guardian linking
Attendance	Ghost students, paper registers, no fraud detection	Teacher-locked, time-windowed (7–9 AM), post-save edit alerts to deputy
Fee Management	Manual ledgers, no real-time visibility, cash leakage	M-Pesa STK Push + C2B, FIFO multi-child split, immutable ledger
Academic Tracking	No early warning, end-of-term-only visibility	CBC + 8-4-4 grading, at-risk flags, trend graphs, report card PDF
Discipline Log	Parents unaware until incidents escalate	Every incident parent-visible same day it is logged
Parent Portal	Parents locked out until report day	Phone OTP login, M-Pesa payment, multi-child switcher, all tabs in one

3.2 Competitive Advantage

Competitor	Gap	SchoolPulse Response
Paper + Excel	No automation, fraud-prone, no analytics	Full digital replacement, same simplicity
Zeraki / uSMART	Expensive, complex UI, enterprise-focus	Cheaper, simpler, pilot-friendly pricing
Elimu SMS	SMS-only, no dashboard or fee management	Full platform: fees + academics + discipline + portal
WhatsApp groups	Unstructured, no records, no delivery tracking	Structured automated alerts with delivery logs

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Business Model & Pricing

Term-based SaaS. SMS on school accounts.

4.1 Pricing Structure

Billing is per term (3 terms per year). Schools think, budget, and pay in terms. Monthly billing invites cancellation between terms. Term-based billing locks the subscription to the school calendar and improves cash flow predictability.

Tier	School Size	Per Term (KES)	Per Year (KES)	Target School Type
Starter	≤400 students	7,500	22,500	Small rural primaries, new private schools
Growth	401–900 students	15,000	45,000	Mid-size primaries, small-to-mid secondaries
Pro	901–1,800 students	26,000	78,000	Large secondaries, big urban primaries
Enterprise	1,800+ students	38,000+	Custom	National schools, county giants, mega primaries

💡 Cost perspective: At KES 7,500/term, a 300-student school pays KES 25 per student per term — less than one pen. The M-Pesa integration alone recovers this cost within the first week of fee collection.

4.2 Additional Revenue Lines

Revenue Line	Rate	Notes
School Setup Fee (one-time)	KES 5,000–10,000	Covers onboarding, data migration, staff training. Filters non-serious schools.
Annual Prepay Discount	–10% off 3-term total	Incentivises full-year lock-in. Improves founder cash flow.
Referral Waiver	1 month free per referral	Schools refer schools. No paid marketing needed in early stage.
SMS Credits (school-owned)	KES 0 to SchoolPulse	Schools top up their own Africa's Talking sub-accounts. Zero SMS cost to platform.

4.3 SMS Infrastructure Model

SMS is one of the highest-risk variable costs in a school management platform. SchoolPulse eliminates this risk entirely through Africa's Talking sub-accounts.

Role	Who Pays	How
Automated SMS (absent, late, fee receipts, reminders)	The school	School tops up their own AT sub-account via M-Pesa. Platform draws from their balance.
Parent OTP login SMS	SchoolPulse (founder)	From master AT account. Estimated KES 2,000–3,000/month across all schools.
Broadcast announcements	The school	Drawn from school's AT sub-account balance.

Sub-account flow:

- SchoolPulse holds a master Africa's Talking account
- Each onboarded school receives a linked sub-account
- School tops up sub-account balance via M-Pesa when low balance alert fires
- All school-triggered SMS draw from that school's own balance
- SchoolPulse never touches school SMS money. Zero liability, zero variable cost exposure.

 SMS volume reality check: A busy 2,000-student primary school can burn 15,000–25,000 SMS/term (attendance + fee reminders + receipts + broadcasts). At KES 0.9/SMS that is KES 13,500–22,500/term — coming from the school's own AT balance, not yours.

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Go-To-Market Strategy

Pilot → referral → partnerships

5.1 Phase 0 — Two-School Pilot (Months 1–2)

Launch with one primary school (1,200–2,000 students) and one secondary school (800–1,200 students). These are not free customers — they are co-development partners. Offer first term free. Provide hands-on onboarding. Sit in the bursar's office on fee day. Watch where it breaks.

Pilot Objective	Success Metric
Validate attendance workflow under real conditions	Register marked in under 3 minutes by all teachers
Validate M-Pesa STK Push end-to-end	At least 20 live parent payments processed
Validate SMS delivery at scale	No missed absent/late alerts during pilot term
Validate parent portal adoption	At least 40% of parents log in within first 2 weeks
Generate reference testimonials	Signed quote from principal and bursar for marketing use

5.2 Phase 1 — Referral-Led Growth (Months 3–12)

School principals and bursars talk at sub-county meetings, workshops, and WhatsApp groups. A bursar who saves 10 hours per week tells the bursar next door. The product demonstrates ROI visually and immediately. Organic referral is the primary channel.

- Offer one free month for every school referred that converts to paid
- Attend sub-county education meetings and TSC forums with a printed one-pager
- Build a short WhatsApp demo video (90 seconds): fee collection → dashboard update → parent SMS receipt
- Target 15–20 paying schools by end of Month 12

5.3 Phase 2 — Channel Partnerships (Year 2)

Partner	Why	Channel
Equity Bank / KCB	Both have school banking products and direct principal relationships	Co-marketing, joint demos at school banking days
Safaricom SME team	M-Pesa integration story is compelling for their SME portfolio	Product partnership, potential bundling

Partner	Why	Channel
KPSA / KSSHA	Institutional endorsement accelerates trust with conservative school heads	Presence at annual conference, endorsement letter
Sub-county education offices	Schools trust their recommendations above commercial vendors	Brief education officers, provide demo accounts

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Financial Projections

Conservative. Term-based. Margin-positive by Month 4.

6.1 Revenue Projections

Period	Paying Schools	Avg Tier	Revenue / Term (KES)	Annual ARR (KES)
Term 2 2026 (Month 4)	2	Growth	30,000	90,000
End of 2026 (Month 9)	10	Growth	150,000	450,000
Mid 2027 (Month 15)	30	Growth/Pro	525,000	1,575,000
End of 2027 (Month 21)	70	Growth/Pro	1,225,000	3,675,000
End of 2028 (Month 33)	200	Mixed	3,500,000	10,500,000
Year 3 (Month 36+)	400	Mixed	7,000,000	21,000,000

6.2 Break-Even Analysis

✓ Break-even is 2 paying schools on Growth tier (KES 30,000/term = KES 10,000/month effective). Monthly operational cost at 1–10 schools is ~KES 7,550. You are profitable before your third paying school.

Schools	Term Revenue (KES)	Monthly Effective (KES)	Monthly Opex (KES)	Net Monthly (KES)
1 school (Growth)	15,000	5,000	7,550	-2,550 (not yet)
2 schools (Growth)	30,000	10,000	7,550	+2,450 ✓
5 schools (Mixed)	87,500	29,167	7,550	+21,617 ✓
20 schools (Mixed)	350,000	116,667	19,550	+97,117 ✓
50 schools (Mixed)	875,000	291,667	35,000	+256,667 ✓

07

Startup Budget & Cost Estimates

Everything you need to launch and survive 12 months

7.1 One-Time Startup Costs

Item	Cost (KES)	Notes
Company name reservation (eCitizen)	150	Lock 'SchoolPulse' immediately. Takes 10 minutes.
Private Limited Company registration	10,650	eCitizen. Do this the moment first school pays.
Domain names (.co.ke + .com)	3,000	schoolpulse.co.ke + schoolpulse.com. ~KES 1,500 each/year.
Professional email (Google Workspace, Year 1)	16,800	KES 1,400/month. Business credibility essential for school principals.
Logo & brand identity	8,000	Fiverr or local designer. One-time. Do not skip this.
Pilot agreement legal template	5,000	One-time. Use a lawyer to draft. Reuse for all pilots.
Pilot SMS credit (2 schools, 1 term)	20,000	You cover pilot school SMS. This is marketing spend, not opex.
Physical marketing materials (one-pagers, folders)	5,000	For school visit demos. Printed, branded.
Transport for school visits (pilot term)	8,000	Matatu + Uber for 10–15 site visits during pilot.
Contingency / miscellaneous	10,000	Buffer for unexpected costs during setup.
TOTAL ONE-TIME	86,600	

7.2 Monthly Operational Costs — Early Stage (1–10 Schools)

Service	Monthly (KES)	Notes
Supabase (free tier)	0	Free up to 500MB DB + 2GB storage. Sufficient for first 10–15 schools.
Railway Starter (backend hosting)	650	~\$5/month. Node.js + Express + Redis.
Vercel (frontend hosting)	0	Free tier more than sufficient for early stage.
Africa's Talking master account (OTP only)	2,500	~2,800 parent OTP logins/month across all schools. Your only SMS cost.

Service	Monthly (KES)	Notes
Google Workspace (email)	1,400	Already counted in one-time Year 1 cost above.
Sentry error monitoring	0	Free tier: 5,000 errors/month. Sufficient for early stage.
Internet + data (development + ops)	3,000	Home fibre or data bundles.
TOTAL EARLY STAGE	7,550	Per month. Covered by 2 paying schools.

7.3 Monthly Operational Costs — Growth Stage (20–60 Schools)

Service	Monthly (KES)	Notes
Supabase Pro	3,250	\$25/month. 8GB DB + 100GB storage. Handles ~60 schools comfortably.
Railway Pro (backend)	3,250	\$25/month. More RAM, auto-scale workers.
Vercel Pro (frontend)	2,600	\$20/month. Analytics + better edge performance.
Africa's Talking master (OTP)	4,500	~5,000 OTP SMS/month across growing school base.
Redis cloud (BullMQ job queues)	1,300	\$10/month. SMS retry queues, fee reminders, scheduled jobs.
Sentry Team	3,250	\$25/month. Error tracking + performance monitoring.
Google Workspace	1,400	KES 1,400/month ongoing.
TOTAL GROWTH STAGE	19,550	Per month. Covered by ~4 schools on Growth tier.

7.4 Monthly Operational Costs — Scale Stage (100–300 Schools)

Service	Monthly (KES)	Notes
PostgreSQL on AWS RDS (db.t3.medium)	13,000	~\$100/month. Migrate from Supabase when hitting DB limits.
Railway / AWS EC2 (backend)	13,000	~\$100/month. Multiple workers, load balancer.
Vercel Enterprise or Cloudflare Pages	6,500	~\$50/month. Full CDN edge, analytics.
Africa's Talking master (OTP)	9,750	~10,000+ OTP SMS/month.
Redis Elasticache / Upstash	3,900	~\$30/month. Queue management at scale.

Service	Monthly (KES)	Notes
Sentry Business	6,500	~\$50/month. Full performance + alerting.
Cloudflare R2 storage (CSV, PDFs)	1,300	~\$10/month. Bulk report storage.
Monitoring (Datadog or similar)	5,200	~\$40/month. Infrastructure dashboards.
TOTAL SCALE STAGE	59,150	Per month. <4% of revenue at 100 schools on Growth tier.

7.5 12-Month Bootstrap Budget Summary

Budget Line	KES	Notes
One-time startup costs	86,600	Registration, branding, legal, pilot SMS, marketing materials
12 months early-stage opex	90,600	KES 7,550 × 12 months
Contingency (15%)	26,580	Buffer for price changes, unexpected infra spikes
TOTAL 12-MONTH BOOTSTRAP BUDGET	203,780	≈ KES 200,000 to launch and survive first year

💡 At 5 paying schools (Month 6–7), monthly revenue of ~KES 58,000 covers all opex + recovers startup costs within 4–5 months of paid operations. The business is self-funding before the end of Year 1.

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Technology Stack

Chosen for speed, scale, and solo-founder survivability

Layer	Technology	Tier	Monthly Cost
Frontend	React.js + Tailwind CSS	Free	KES 0
Mobile (Phase 2)	React Native	Free	KES 0
Backend	Node.js + Express.js	Free	KES 0
Database	PostgreSQL via Supabase	Free → Pro	KES 0 → 3,250
Authentication	JWT + Africa's Talking OTP	Usage-based	~KES 2,500
Payments	Safaricom Daraja API (STK Push + C2B)	Free (go-live needs biz docs)	KES 0
SMS Delivery	Africa's Talking sub-accounts (school-owned)	School pays	KES 0 to platform
PDF Generation	Puppeteer (HTML→PDF) + pdf-lib merge	Free	KES 0
Background Jobs	BullMQ + Redis	Free → KES 1,300/month	KES 0 → 1,300
File Storage	Supabase Storage → Cloudflare R2	Free → KES 1,300/month	KES 0 → 1,300
Frontend Hosting	Vercel	Free → Pro	KES 0 → 2,600
Backend Hosting	Railway	Starter → Pro	KES 650 → 3,250
Error Monitoring	Sentry	Free → Team	KES 0 → 3,250
WhatsApp (Phase 2)	Meta Cloud API	1,000 free msgs/month	KES 0 → usage

Why Supabase over raw PostgreSQL

- Free tier handles first 10–15 schools: 500MB DB, 2GB storage, 50,000 monthly active users
- Built-in auth helpers reduce JWT implementation time
- Built-in storage replaces separate S3 setup
- Real-time subscriptions available for dashboard live updates
- One-click migration to dedicated PostgreSQL when outgrowing free tier

Why BullMQ for background jobs

- SMS retry logic (up to 3 retries over 30 minutes if AT delivery fails)
- Scheduled fee reminders (daily, weekly, bi-weekly per school config)
- Webhook processing queue (M-Pesa C2B payloads persisted before processing)
- Report card batch PDF generation (runs async, doesn't block UI)

09

Company Registration & Legal

When to register, what structure, can you sell before?

9.1 Selling Before Official Registration — Yes, Legally Possible

In Kenya, you can sign contracts and receive payment as an individual before incorporating a company. A Pilot Agreement or Letter of Intent signed by you personally is legally valid. However, there are important rules:

Action	Before Registration	After Registration
Sign pilot agreement	✓ Valid personally	✓ Under company name (better)
Receive payment (M-Pesa / bank)	✓ Personal account temporarily	✓ Company account (required for tax)
Issue receipt / invoice	✓ As individual	✓ Under company + KRA PIN
Claim data processor status (DPA 2019)	⚠ Uncertain	✓ Clear legal standing
Raise investment / pitch formally	✗ Very difficult	✓ Clean cap table

⚠ Register the company the moment the first school pays — not after. Data protection liability under the Kenya DPA 2019 attaches to whoever signed the data processing agreement. You want a company between you and that liability.

9.2 Registration Steps & Costs

Step	Action	Cost (KES)	Timeline
1	Reserve company name on eCitizen (do this NOW)	150	10 minutes
2	Register Private Limited Company (eCitizen)	10,650	3–7 business days
3	Apply for KRA PIN (company)	0	Same day
4	Open company bank account (Equity / KCB)	0–5,000	1–3 days
5	Register for VAT (once turnover hits KES 5M/year)	0	When applicable
6	Name reservation (do today, locks 'SchoolPulse')	150	Immediate

9.3 Intellectual Property

- Register SchoolPulse® as a trademark with Kenya Industrial Property Institute (KIPI). Cost ~KES 8,000. Do this before public launch.
- All source code is proprietary. Add a LICENSE file to the private repository explicitly. Do not use GPL-licensed dependencies.
- All school data is owned by the school, not SchoolPulse. Your data processing agreement must state this explicitly — required by Kenya DPA 2019.

10

Team & Hiring Plan

Solo now. Strategic hires at the right milestones.

10.1 Founder

Name	Role	Skills	Contact
Dennis Wambugu	Founder & Lead Engineer	React.js, Node.js, Java, JavaFX, PostgreSQL, full-stack product development	mainadenniswambugu@gmail.com

10.2 Hiring Milestones

Trigger	Hire	Compensation Model
Month 4–5: 5 paying schools	Second full-stack developer (React + Node)	Equity-based or KES 50,000–80,000/month once revenue covers
Month 8–10: 20 schools	Sales / school relations executive	Commission (10–15% of closed ARR) + equity upside
Month 12+: 40 schools	UI/UX designer (part-time → full-time)	Project-based initially, then full-time
Year 2: 100 schools	Customer success manager	KES 60,000–80,000/month. Handles onboarding + retention.

10.3 Advisory Needs

- Education sector advisor — MOE or KPSA/KSSHA connections for institutional access
- Safaricom / fintech advisor — to accelerate Daraja go-live and potential Safaricom partnership
- Legal advisor — Kenya DPA 2019 compliance, KIPI trademark registration, investor agreements

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Risks & Mitigations

Honest. Actionable.

Risk	Likelihood	Impact	Mitigation
Schools resist change from paper	High	Slow adoption	Free pilot + personal onboarding + show time-saved in Week 1
Unreliable internet at school	High (rural)	Feature gaps	PWA offline mode: attendance works offline, syncs on reconnect
Daraja M-Pesa go-live delay	Medium	Delayed fee feature	Launch with manual fee recording first. M-Pesa is enhancement, not blocker.
School cancels after pilot	Low–Medium	Zero revenue	Setup fee filters low-intent schools. Data lock-in builds switching cost.
Competitor copies product	Medium (Year 2+)	Market share	Kenya-first depth + historical data + school relationships = the moat
Data breach / privacy incident	Low	Legal + trust	TLS, AES-256, JWT, KDPA compliance, no hard deletes, audit logs everywhere
School can't pay subscription	Medium	Revenue loss	Term billing + auto-downgrade to read-only. Data never deleted.
Solo developer bottleneck	High (early)	Slow delivery	Hire second developer at Month 4–5. This plan funds that hire.
Africa's Talking outage	Low	SMS disruption	Queue + retry (BullMQ). Attendance saves regardless. SMS queued for retry.

12

Exit Strategy

Build it. Scale it. Sell it right.

SchoolPulse is designed to be built, scaled, and sold. The founder does not intend to operate it indefinitely. The exit thesis is built into the product architecture from Day 1.

12.1 Acquisition Targets

Acquirer Type	Why They'd Buy	Estimated Timeline
Safaricom	SchoolPulse is a fee collection machine built on M-Pesa. They'd acquire distribution and school relationships.	Year 3–4 at 200+ schools
Equity Bank / KCB	Both have school banking units. SchoolPulse is ready-made school financial infrastructure.	Year 2–3 at 100+ schools
Larger EdTech (regional)	East Africa expansion without building from scratch. Proven product, school relationships, KDPA compliant.	Year 3–5
PE / Strategic investor	Take SchoolPulse regional (Uganda, Tanzania, Rwanda) at proven Kenyan unit economics.	Year 4–5 at 400+ schools

12.2 Valuation Levers

- ARR multiple: SaaS businesses in Africa typically exit at 3–6× ARR. At KES 42M ARR (Year 3) → KES 126M–252M valuation.
- Strategic premium: M-Pesa integration + school relationships + KDPA compliance adds strategic value above pure ARR multiple.
- Retention rate: High switching cost (historical data, parent accounts, staff training) = high NRR. Buyers pay for this.
- Data asset: Attendance, academic, and financial data across hundreds of schools is valuable for EdTech analytics buyers.

12.3 Pre-Exit Preparation

- Maintain clean cap table from Day 1. Use a company from first payment.
- Document all code, architecture, and processes. Bus factor of 1 kills acquisition value.
- Hire a second developer before raising. Acquirers discount solo-founder engineering risk heavily.
- Get audited accounts from Year 2 onwards. Non-negotiable for any serious acquisition conversation.

 The goal: reach 200–400 schools, prove unit economics, and run a competitive acquisition process. Build it right. Build it clean. Sell it well.

— *End of Business Plan* —

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